BI X

Digital Innovation Lab by
Boehringer Ingelheim

Research Readout

A/B Testing – SurvoPlus Prototype

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Agenda

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2 Study Design
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4 Comparative Analysis: Control vs AI

5 Emerging Themes and Strategic Risks

6 Recommendations & Next Steps

SurvoPlus is a patient support app that uses AI to deliver highly personalized insights and content.

Adaptive UI: Personalizes features and content based on each patient's journey, goals, and context.

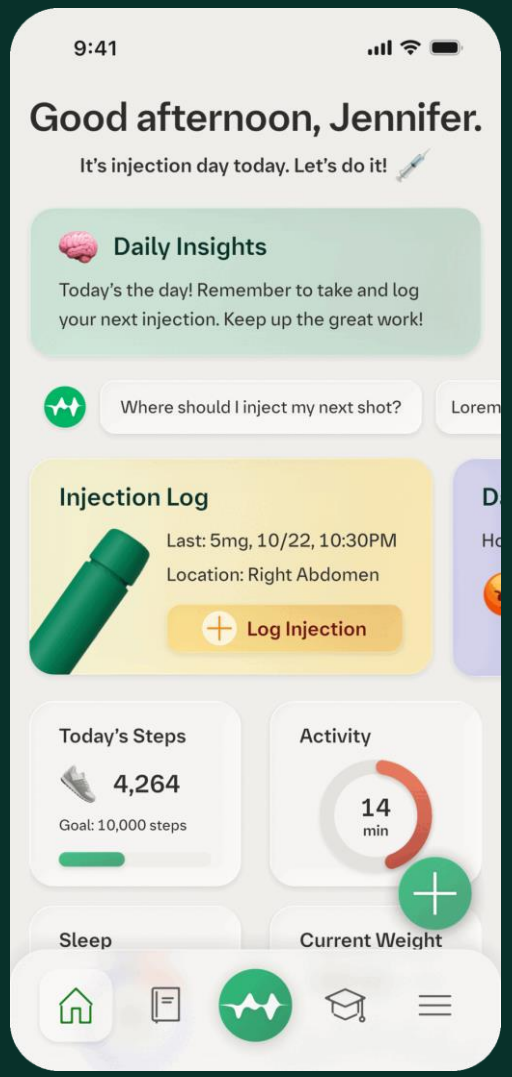
AI Assistant: Offers tailored support, education, and encouragement, adapting its responses to the patient's needs.

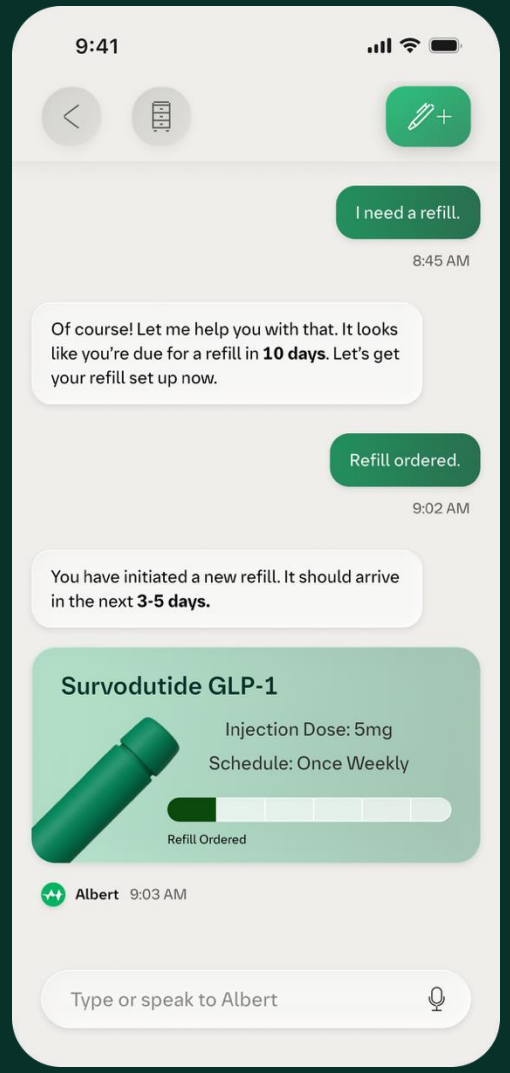
Our goals:

→ Improved adherence

→ Better patient outcomes

→ Lasting behavioral change





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Executive Summary

Executive Summary

Context

In December 2025, BI X conducted **unmoderated split tests** (i.e., A/B testing) with **17 GLP-1 patients** to validate whether the AI-driven features in SurvoPlus deliver **measurable value** compared to a more traditional, non-AI control version. Two versions of the MVP were tested:

- **Control Version**: Standard logging-based app experience without AI enhancements.
- **AI Version**: Core app experience + AI chat, adaptive UI, and AI-assisted food logging.

This testing served as the **conclusion and validation of our MVP phase.**

Strong Preference for the AI Version over the Control

AI group participants reported **significantly** higher satisfaction scores (avg. **4.6/5** vs **3.2/5** for Control) and described the app as more helpful and engaging.

8/11 of the AI participants indicated a **willingness to pay for the app**, and a majority indicated that it **would help them stay adherent**.

Personalization Enhanced User Confidence

Participants highlighted **tailored recommendations** and **contextual prompts** as critical for reducing friction and improving adherence. **73% of AI group users cited personalization as a key driver of satisfaction.**

> I would use it because it's personalized to me. *(GLG_04, Survey Comment)*

Nutrition & Meal Logging Emerged as a Core Value Driver

While initially scoped as a supporting feature, **meal logging proved to be a top priority for participants**, far exceeding expectations.

Users emphasized the convenience of AI-powered photo analysis and expressed frustration with manual entry in the Control group.

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Study Design Participant Profile

Study Design Overview

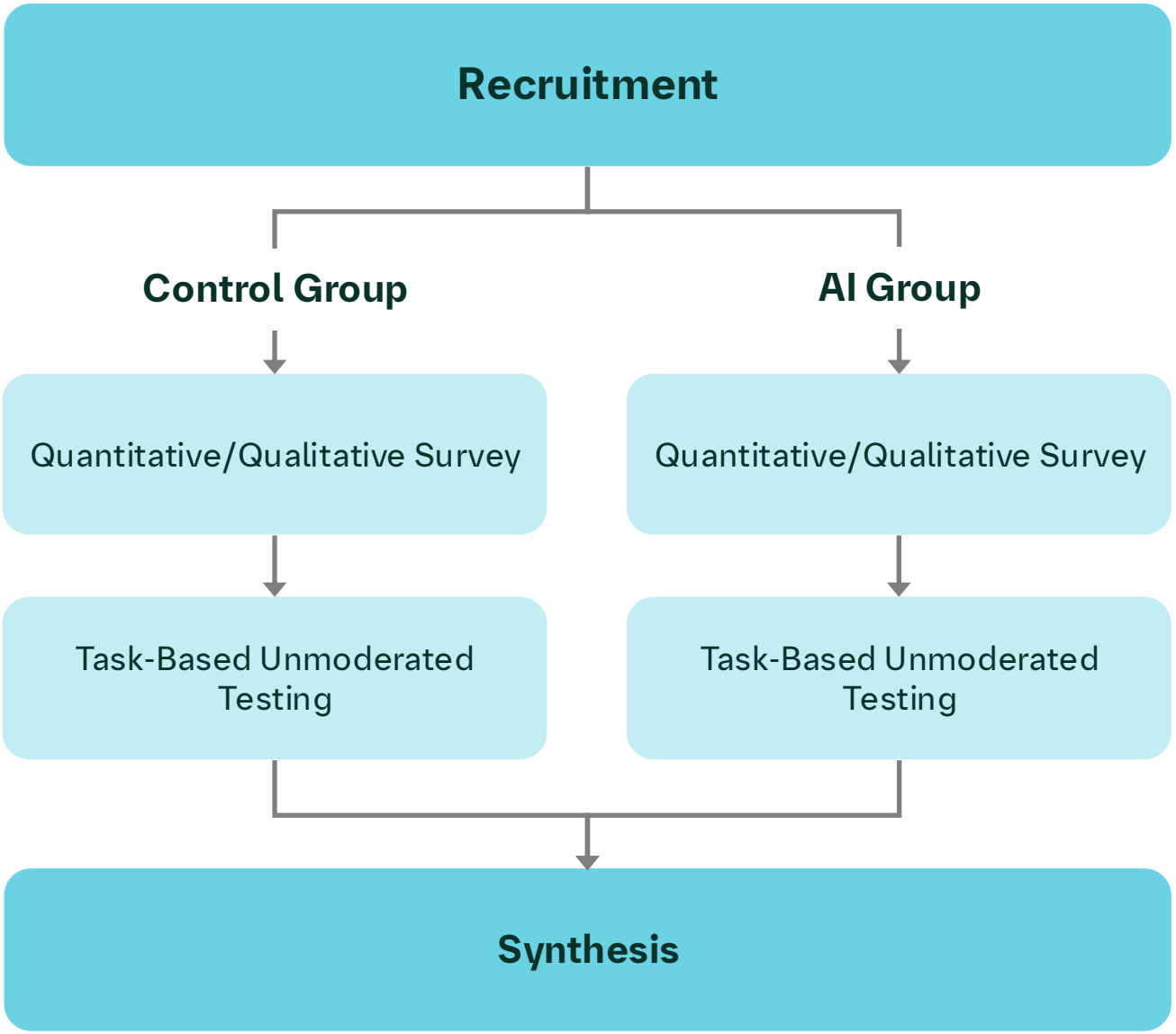
A/B Testing Approach - Two interactive web app prototypes tested:

- **Control Version**: Standard app experience without AI enhancements.
- **AI Version**: Adaptive UI, AI-assisted food logging, and "Albert" chat feature.

Testing Methodology - Combination of structured surveys and unmoderated sessions via Lookback.

Recruitment - Managed through GLG; participants met strict inclusion criteria:

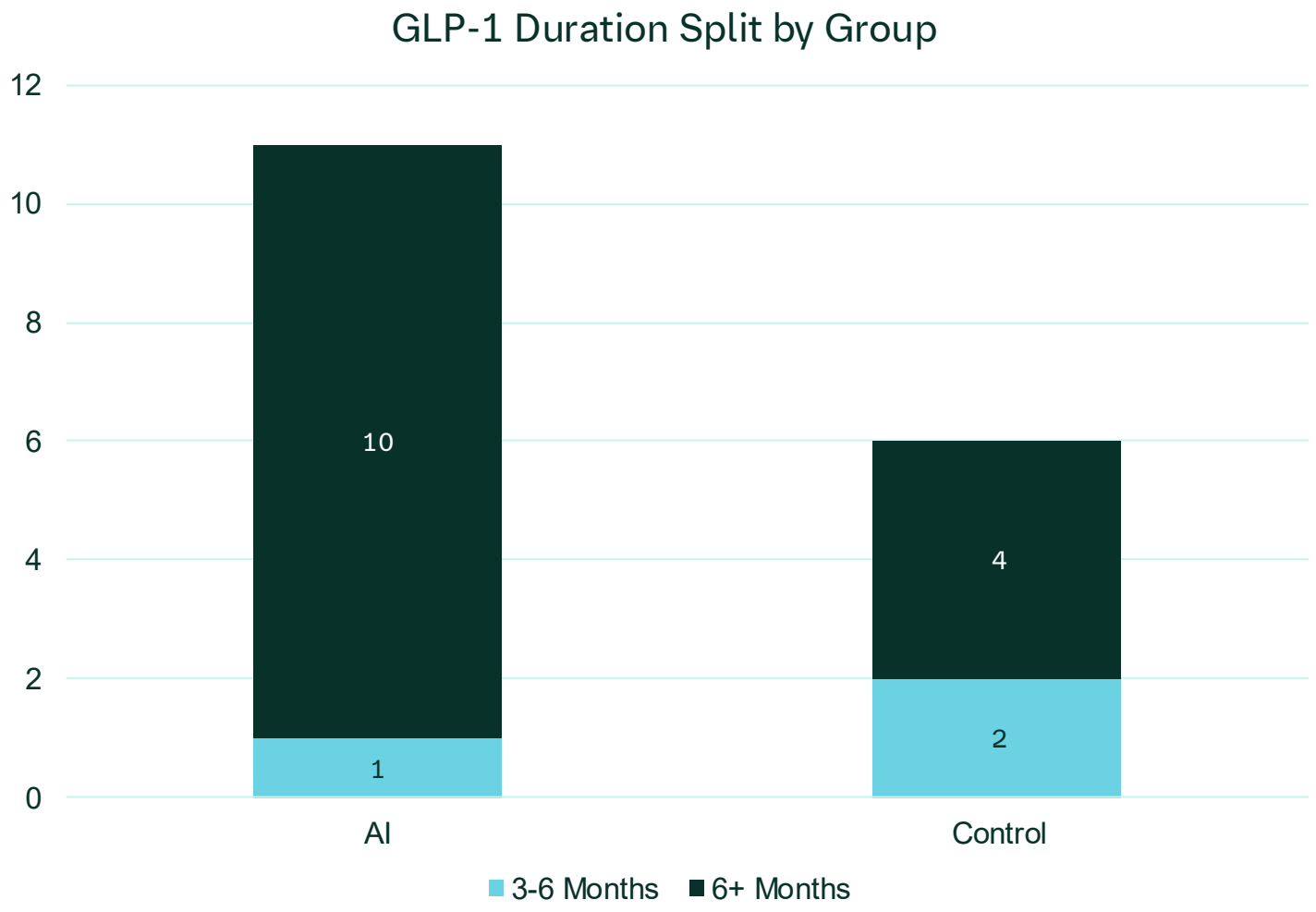
- Age: 20–60 years
- Currently using GLP-1 therapy
- On drug for 3–6 months or 6+ months
- Diagnosed with obesity
- Active/recent users of digital health platforms



Participant Counts & Cohort Balance

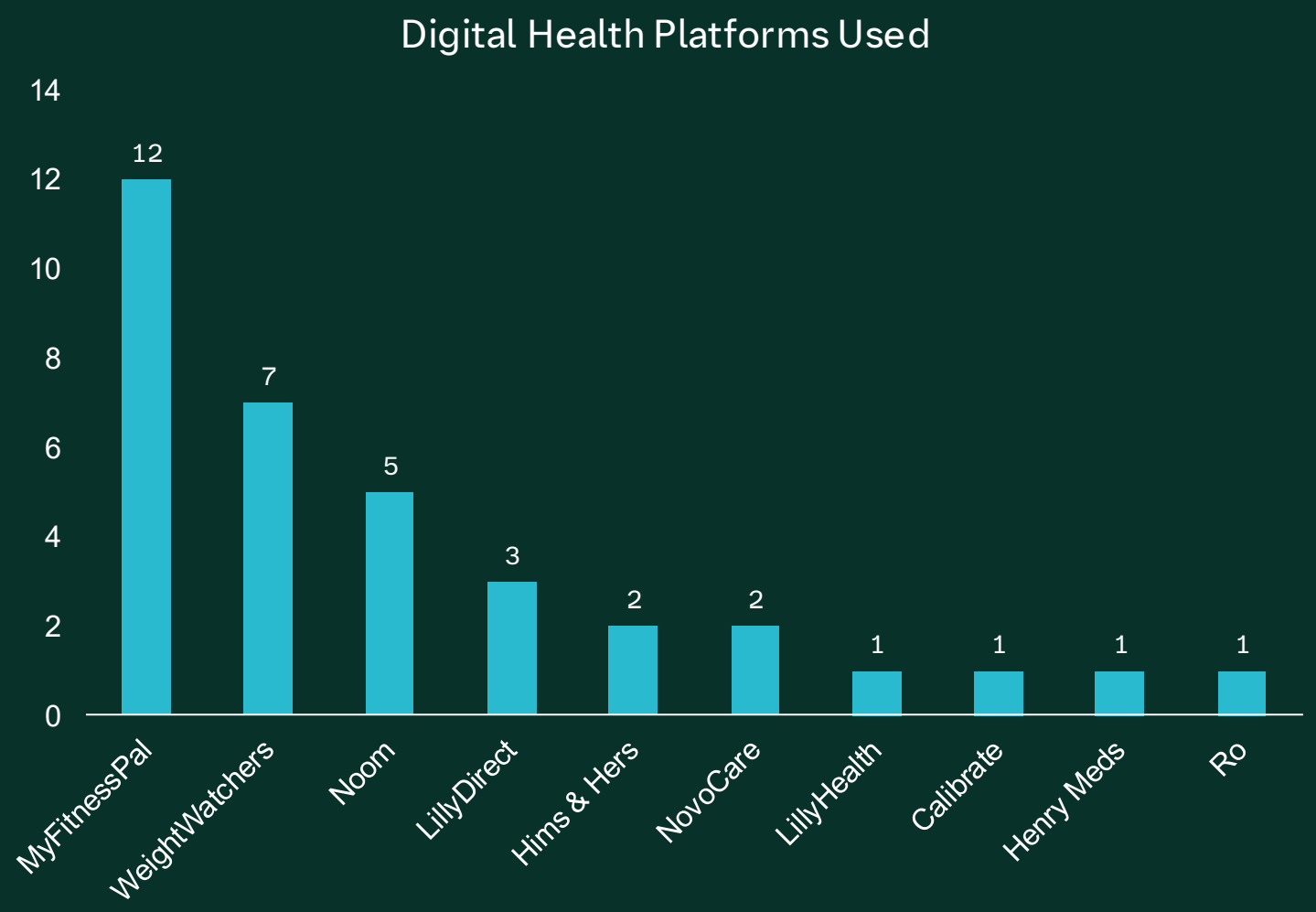
- Initially recruited: **24 participants**
- Completed study: **17 participants**
  - AI Group: **11 participants**
  - Control Group: **6 participants** **
- Cohorts balanced for age, gender, GLP-1 duration, and digital health app / PSP experience.

** Although the Control group had a lower overall participation rate than the AI Group, research from the Nielsen Norman Group shows this sample size is sufficient for meaningful feedback.

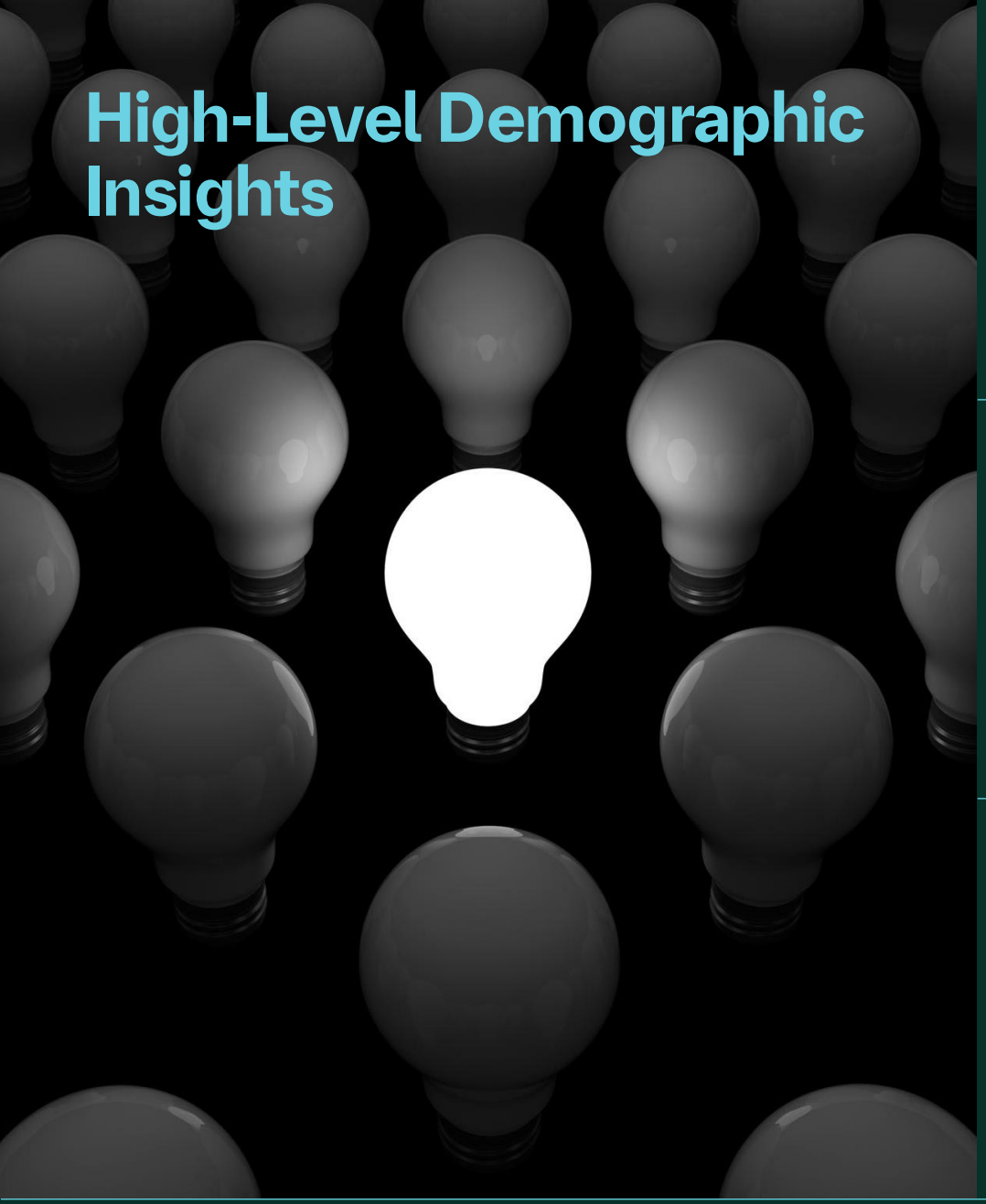


Participant Demographics

- **Age Range**: 32–47 years (median ~40)
- **Gender**: ~70% female
- **GLP-1 Duration**:
  - 3–6 months: ~18%
  - 6+ months: ~82%
- **Most Common Digital Health Platforms Used**: MyFitnessPal, WeightWatchers, Hims & Hers, NovoCare



High-Level Demographic Insights



Participants skewed toward **tech-savvy individuals** familiar with digital health tools.

Strong representation of **long-term GLP-1 users (6+ months)** suggests readiness for advanced features.

High engagement with **nutrition-focused platforms** aligns with strong interest in meal logging and personalized guidance.

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Research Objectives and Findings

Research Objectives

1 Perceived Value of AI Features

2 Effectiveness of Adaptive UI & Conversational Interface

3 Differentiation vs Other Digital Health Solutions

4 Impact on Adherence, Retention, and Behavior Change

Objective 1:

Perceived Value of AI Features

Determine whether AI-driven capabilities (adaptive UI, AI chat assistant "Albert," photo-based meal logging) deliver **greater perceived value** compared to the non-AI Control version.

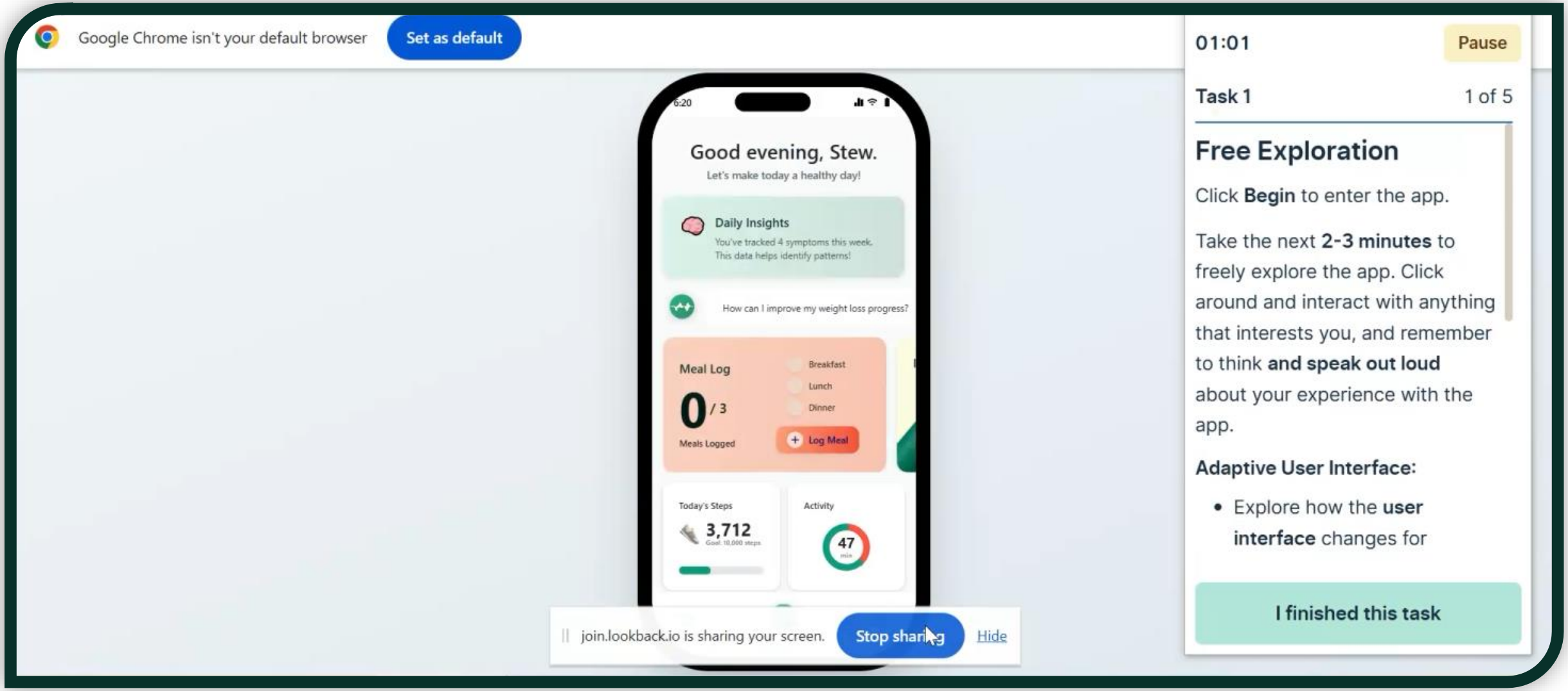
Context & Summary

Participants Strongly Preferred the AI App Version

Participants **overwhelmingly favored the AI-enhanced app experience**, citing convenience, personalization, and proactive support as key differentiators.

> “Albert is really cool, I have to say.”
>
> *- (GLG_21, Session Transcript)*

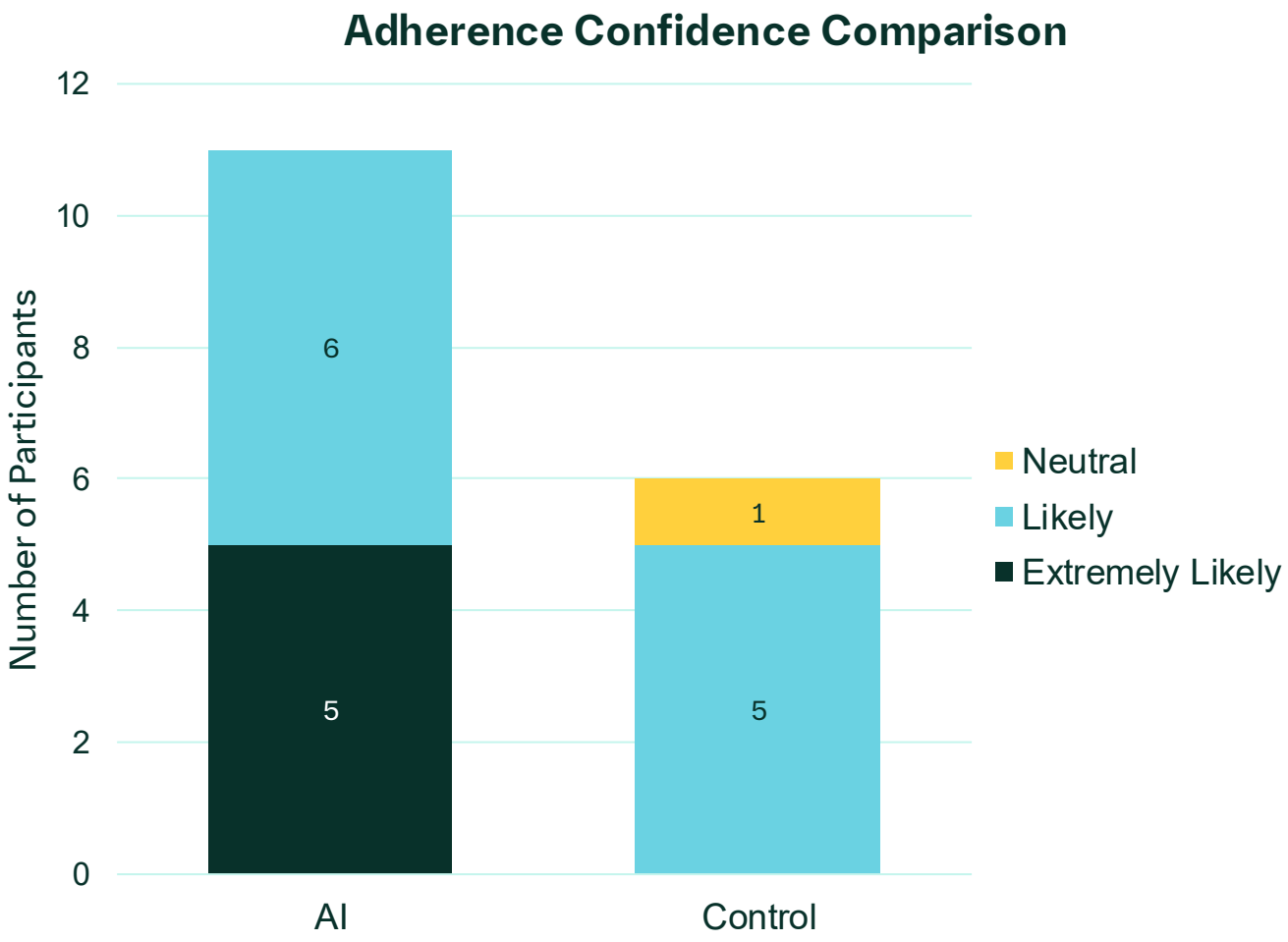
The **AI chat assistant emerged as the most impactful feature**, providing quick, relevant guidance and reducing friction in managing GLP-1 therapy. Photo-based meal logging was also highly valued, transforming a previously tedious task into a seamless experience.


Google Chrome isn't your default browser
Set as default
Good evening, Stew.
Let's make today a healthy day!
Daily Insights
You've tracked 4 symptoms this week.
This data helps identify patterns!
How can I improve my weight loss progress?
Meal Log
0 / 3
Meals Logged
Breakfast
Lunch
Dinner
Log Meal
Today's Steps
3,712
Activity
47
join.lookback.io is sharing your screen.
Stop sharing
Hide
01:01
Pause
Task 1
1 of 5
Free Exploration
Click Begin to enter the app.
Take the next 2-3 minutes to freely explore the app. Click around and interact with anything that interests you, and remember to think and speak out loud about your experience with the app.
Adaptive User Interface:
• Explore how the user interface changes for
I finished this task

Quantitative Analysis

Key Metrics

- **Satisfaction Scores:**
  - **AI - 4.6/5**
  - **Control - 3.2/5**
- **Preference Rate:**
  - AI: **91%**
  - Control: **33%**
- **Willingness to Pay:**
  - AI: **8 of 11**
  - Control: **1 of 6**
- **Adherence Confidence:**
  - AI: majority rated “likely to improve adherence”
  - Control: minority



Qualitative Analysis

Themes

- **AI Chat as Hero Feature:** Participants praised Albert for being **fast, helpful**, and surprisingly intuitive.
- **Meal Logging Convenience:** Photo-based logging repeatedly called a **"game-changer."**
- **Personalization:** Users **valued tailored recommendations** over generic advice.
- **Control Group Frustration:** Lack of personalization and manual entry led some to explicitly suggest adding AI features.

What AI Group Said:

> I really like how it gives like things that you may want to ask AI . . . having some prompts that you can click kind of gets that ball rolling which I think is really cool.

- (GLG_17, Session Transcript)

> If AI can help recognize what food is in your plate and the portion . . .That's wonderful.

- (GLG_25, Session Transcript)

What Control Group Said

> It feels like there's a missed opportunity here re: integrating AI.

- (GLG_14, Survey Comment)

Implications & Recommendations

What This Means

AI features significantly enhance perceived value, validating MVP hypotheses.

Recommendations

- Prioritize refinement of AI chat and inclusion of more embedded/inline features
- Expand meal logging and recipe functionality

Objective 2:

Effectiveness of Adaptive UI & Conversational Interface

Assess whether the AI-driven conversational interface (“Albert”) and adaptive UI elements **improved usability**, **task completion**, and **overall user experience** compared to the Control version.

Context & Summary

AI Assistant received high praise, Adaptive UI emphasizes personalization

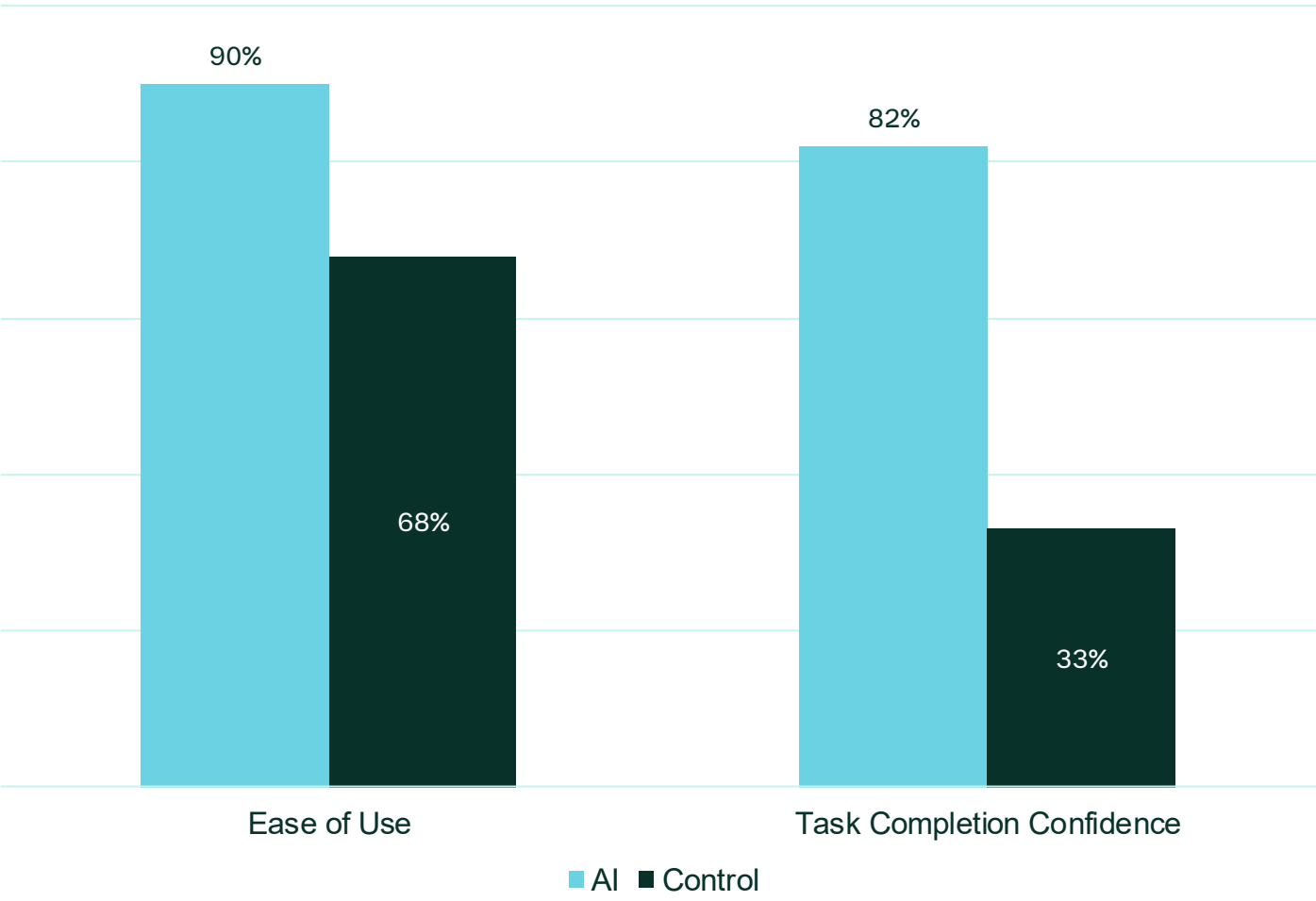
Participants consistently praised the **AI chat assistant ("Albert")** for its ability to **provide quick, relevant answers** and proactive guidance, reducing friction in completing tasks and accessing information. The conversational interface was described as intuitive and confidence-building, especially for complex queries like recipe suggestions and symptom management.

The **adaptive UI**, which tailored content and navigation based on user context, was positively received for making the experience feel **streamlined and personalized**. However, several participants noted that they didn't notice adaptive changes or felt the personalization wasn't strong enough, signaling **an opportunity to make adaptive elements more visible and impactful**.

Quantitative Analysis

Key Metrics

- **Ease of Use Ratings:**
  - AI: **4.5/5**
  - Control: **3.4/5**
- **Task Completion Confidence:**
  - AI: **82% very confident**
  - Control: **33%**
- **Feature Usage:**
  - AI chat used by **100% of AI participants**
  - Adaptive UI features (persona switching, contextual prompts) noted by **73%**
- **Control Group Contrast:**
  - Manual navigation and lack of guidance cited as top usability pain points.



Qualitative Analysis

Themes

- **Conversational Interface:** Albert was described as **“fast,” “clear,”** and “like having a personal guide.”
- **Adaptive UI:** While appreciated, some participants said they **didn’t notice adaptive changes or wanted stronger personalization** and visual differentiation.
- **Control Group Frustration:** Users struggled with static navigation and generic content, explicitly requesting “smarter” features.

What AI Group Said:

> You may even change themes for different progress.

- (GLG_25, Survey Comment)

What Control Group Said

> I wish there was more customizability. Everything felt quite generic and generalized.

- (GLG_14, Survey Comment)

Implications & Recommendations

What This Means

Conversational interface is the primary driver of usability improvements; adaptive UI adds value but lacks visibility and depth.

Recommendations

- Enhance Albert's capabilities (contextual depth, proactive prompts).
- Expand adaptive UI personalization and **make adaptive changes visually distinct** (e.g., dynamic layouts, clear indicators).

Strategic Note: Position **conversational interface** as the **core differentiator** while **elevating adaptive UI** to reinforce personalization.

Objective 3:

Differentiation vs Other Digital Health Solutions

Assess whether SurvoPlus –particularly its AI-driven features – offers **clear differentiation** compared to other digital health platforms participants have used (e.g., MyFitnessPal, Noom, WeightWatchers, Calibrate).

Context & Summary

Patients Saw AI Chat as Extremely Differentiating

Participants widely recognized SurvoPlus as **distinct from traditional digital health tools**, primarily due to its **AI-driven conversational interface and integrated GLP-1 support**.

> I really like this [app] because it is so specific to those using GLP-1s.

- (GLG_22, Survey Comment)

Unlike generic calorie trackers or static educational apps, SurvoPlus provided contextual **guidance, proactive prompts, and medication-specific resources**, which participants described as “next-level.”

However, some features – such as **progress tracking and recipe suggestions** – felt similar to existing platforms, indicating an **opportunity to elevate these areas through AI-powered enhancements** like dynamic meal planning and adaptive fitness plans.

Quantitative Analysis

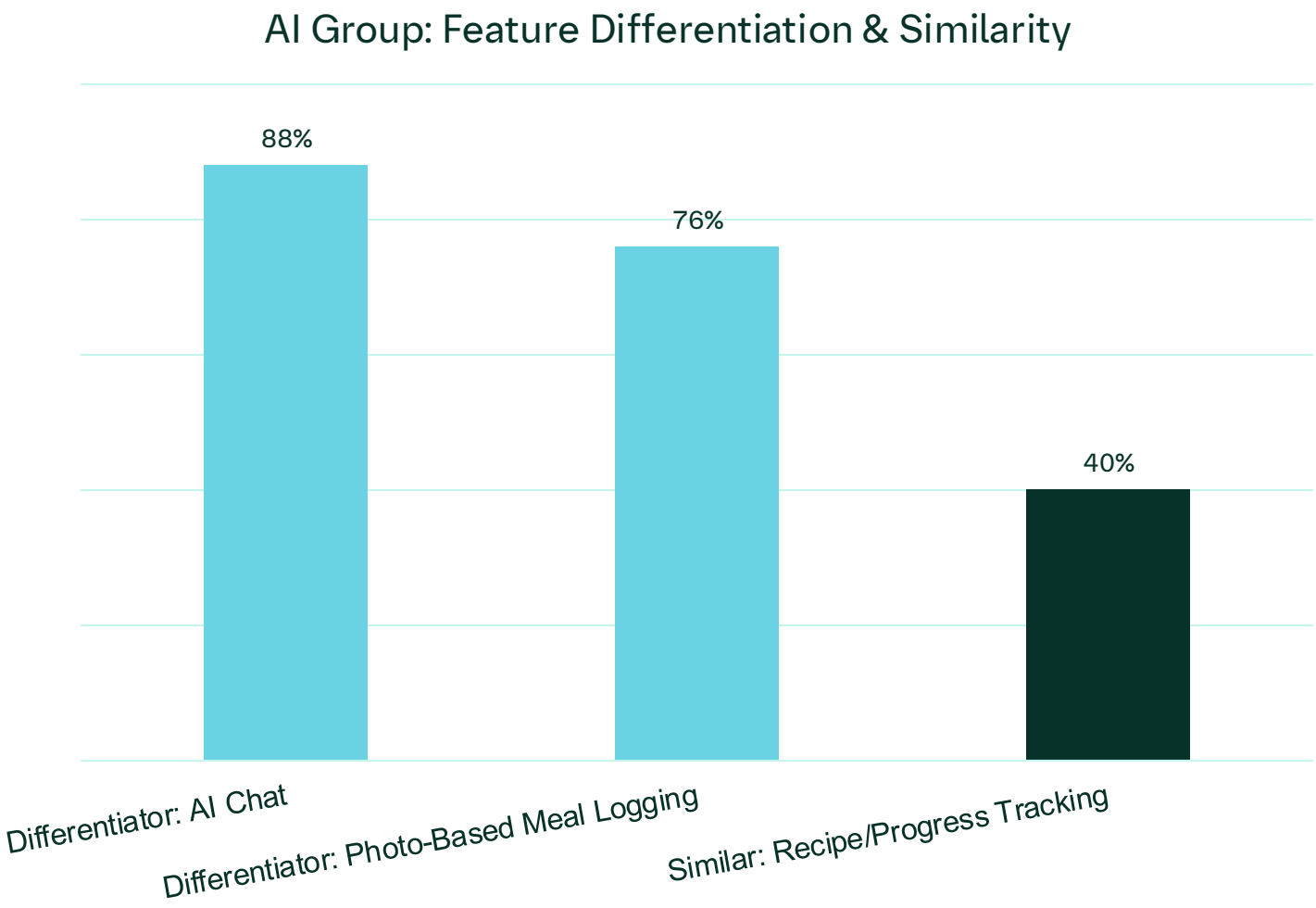
Key Metrics

AI Group:

- Perceived Differentiation: **76%** "significantly more advanced"
- Top Differentiators:
  - AI chat: **88%**
  - Photo-based meal logging: **76%**
  - Overlap Features: **~40%** flagged recipes and progress tracking as "similar"

Control Group:

- Only 33% felt the app was "different enough" from existing solutions.



Qualitative Analysis

Themes

- **AI as Key Differentiator**: Participants emphasized that Albert & photo logging set SurvoPlus apart from “generic trackers.”
- **Clinical Integration**: GLP-1-specific prompts and symptom guidance seen as unique and valuable.
- **Overlap Concerns**: Recipe suggestions and progress graphs felt “like Noom or MyFitnessPal.”
- **Opportunity for AI Expansion**: Participants expressed interest in AI-driven meal planning, shopping lists, and adaptive educational content.

What AI Group Said:

> This app looks great. It’s got everything in one place and it looks like I can ask questions about my progress.

- (GLG_08, Session Transcript)

> I definitely would prefer this app. I love the AI chat feature.

- (GLG_27, Survey Comment)

What Control Group Said

> I definitely prefer LoseIt because it has a database of foods, bar code scanners, etc, to streamline the process of calorie logging. Likewise, FitBit does a better job of activity tracking.

- (GLG_14, Survey Comment)

Implications & Recommendations

What This Means

SurvoPlus differentiates strongly through AI and GLP-1 integration, but some features risk blending into the competitive landscape.

Recommendations

- Double down on **AI-driven personalization and clinical depth** to reinforce uniqueness.
- Elevate overlapping features through **AI-powered enhancements**:
  - Dynamic meal planning tied to photo-based logging.
  - Personalized shopping lists and adaptive fitness plans.
  - Contextual, adaptive educational content.

Objective 4:

Impact on Adherence, Retention, and Behavior Change

Evaluate whether AI-driven features **positively influence adherence** to GLP-1 therapy, retention intentions, and behavior change compared to the Control version.

Context & Summary

AI App Version Demonstrates Higher Adherence and Retention Potential

Participants in the AI group demonstrated **notably higher adherence confidence** and **stronger retention intent** compared to the Control group.

These gains were attributed to **proactive reminders**, **personalized prompts**, and **conversational support**, which reduced friction and encouraged consistent engagement. Behavior change indicators also surfaced within the AI group, with participants showing more interest in meal logging and greater confidence in their ability to maintain routines.

Quantitative Analysis

Key Metrics

Adherence Confidence:

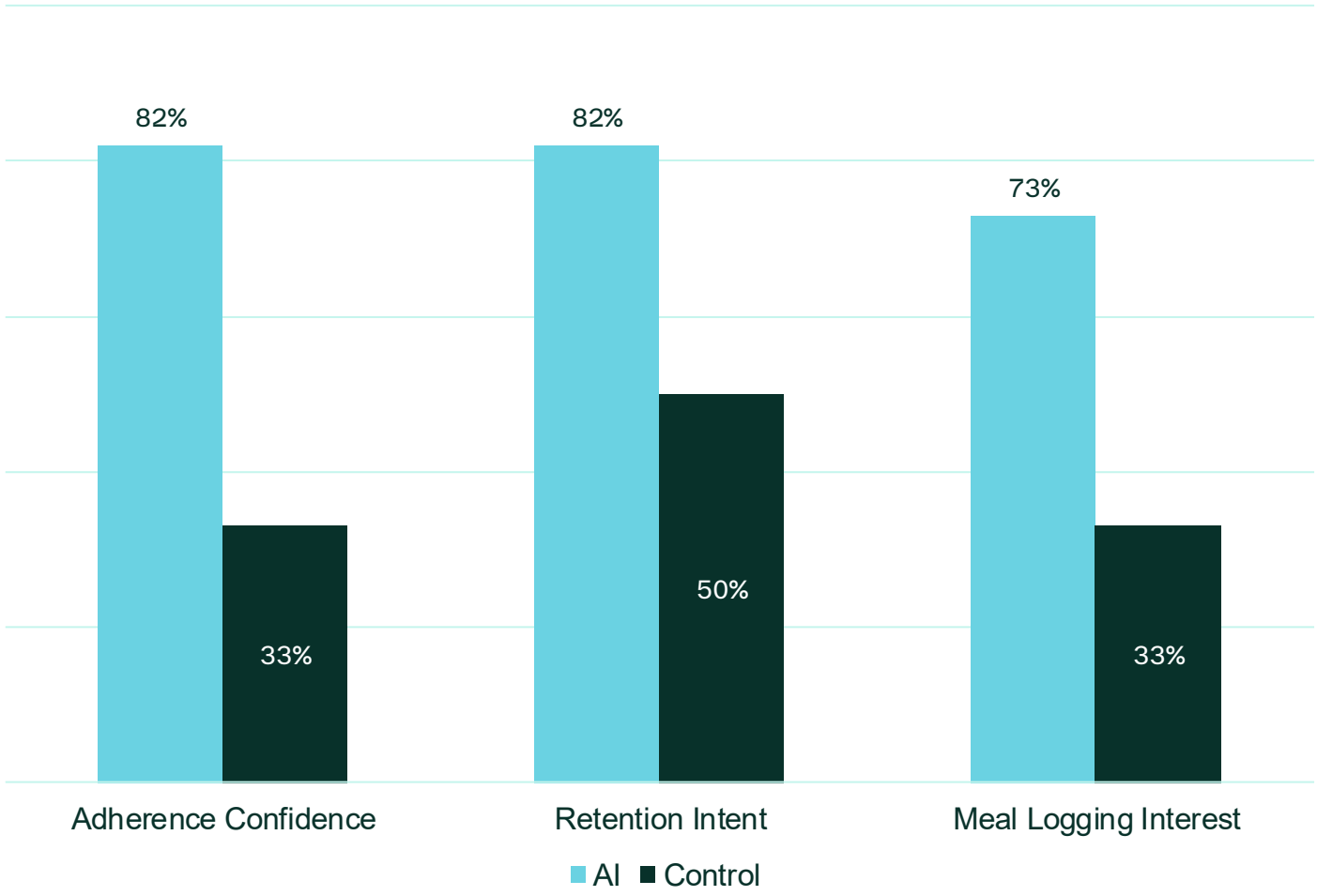
- AI: **82%** reported “very confident”
- Control: **33%**

Retention Intent:

- AI: **9** of **11** participants
- Control: **3** of **6** participants

Interest in Meal Logging:

- AI: **73%**
- Control: **33%**



Qualitative Analysis

Themes

- **Proactive Guidance:** AI prompts and reminders were described as motivating and confidence-building.
- **Conversational Support:** Albert's contextual advice helped participants overcome plateaus and maintain routines.
- **Control Group Frustration:** Lack of reminders and manual logging led to disengagement.

What AI Group Said:

> I would use this app daily and for all my weight loss needs. It appears to have everything in one place.

- (GLG_08, Survey Comment)

What Control Group Said

> My issue is I'll forget. After forgetting so many times I'll eventually stop using the app all together.

- (GLG_16, Survey Comment)

Implications & Recommendations

What This Means

AI features **significantly improve** adherence confidence and retention intent, validating their role in behavior change.

Recommendations

- Enhance proactive guidance (e.g., dynamic reminders tied to user patterns).
- Expand conversational depth for adherence coaching and motivational nudges.
- Integrate **adaptive goal-setting** to reinforce long-term engagement.
- **Include CBT methodology in next phase** by consulting behavioral specialists to strengthen adherence coaching strategies.

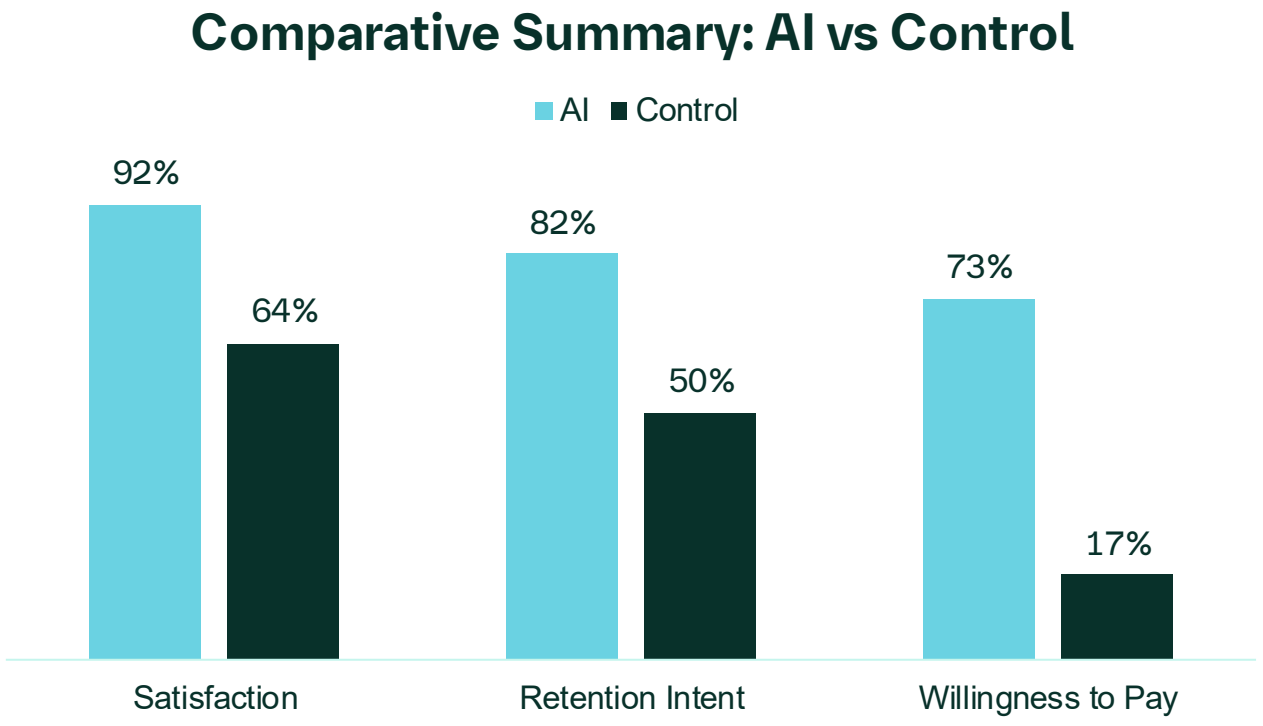
4

Comparative Analysis: Control vs AI

Comparative Summary

Value of AI Features Clearly Demonstrated

Across all objectives, **the AI cohort consistently outperformed the Control group** on satisfaction, engagement, and perceived value. These differences validate the strategic impact of AI-driven features and reinforce their role as core differentiators.



Summary

- Overall Satisfaction: AI 4.6/5 vs Control 3.2/5
- Retention Intent: AI 82% likely to continue vs Control 50%
- Willingness to Pay: AI 73% positive vs Control 17%

Strategic Callbacks – What this Means Overall

Control Group

The control group explicitly requested AI features. This validates the MVP direction.

AI Group

The AI group showed a strong willingness to pay. This positions the AI version as a premium offering.

Cross-Cohort Theme

Both cohorts emphasized they want personalization. This will be critical for the next phase.

5

Emerging Themes and Strategic Risks

Emerging Themes and Strategic Risks

Context

While our core objectives confirm the value of AI-driven features, deeper analysis of qualitative feedback reveals **critical themes that go beyond usability and satisfaction scores**. These insights highlight **emotional drivers, trust gaps, and systemic usability issues** that could significantly influence adoption and retention.

The underlying needs and concerns raised here, if unaddressed, **could undermine SurvoPlus's differentiation**. These themes represent strategic **opportunities** and **risks** as we move into the Pilot phase.

Why This Matters

Addressing these areas during the BI X exploration will:

- Strengthen user trust and emotional safety.
- Reduce friction and improve engagement.
- Position SurvoPlus as a **holistic, empathetic solution**, not just a functional tool.

Mental Health & Emotional UX

Participants repeatedly flagged **daily weight tracking** as emotionally harmful. Even minor fluctuations triggered feelings of failure, especially for those with a history of eating disorders. This isn't just a UX tweak—it's a **mental health safeguard** that impacts trust and long-term engagement.

Data Insights

- 3 participants explicitly raised concerns about daily weight checks.
- Weekly snapshots suggested as a safer alternative.

What Participants Said

"I don't like the daily check-in. Seeing backwards progress would make me feel like I'm backsliding" (GLG_29 – Session Transcript)

Recommendation

- Shift to **weekly weight tracking**.
- Incorporate **CBT-informed prompts** to reduce anxiety.

AI Skepticism and Trust Gap

While participants praised AI features, they also expressed **serious doubts about accuracy and authenticity**. Users questioned if photo-based meal analysis could accurately detect nutrients, and whether Albert's responses were too generic to be trusted. This trust gap is a critical barrier to adoption.

Data Insights

- Multiple AI group participants raised concerns about photo-based meal logging accuracy.
- Trust-building emerged as a recurring theme in survey comments.

What Participants Said

"I do enjoy the app, but I find it unlikely that a photo and AI will be able to tell that I used 1 tablespoon of olive oil while cooking my chicken. It seems improbable."
(GLG_26 – Survey Comment)

Recommendation

- Display **ingredient detection details** and confidence scores.
- Provide **clear explanations of AI logic** and limitations.

Adaptive UI Visibility

Personalization was a core promise—but many participants **didn't notice the adaptive changes** that accompanied different personas.

Subtle UI shifts failed to convey value, leaving users feeling the experience was generic.

Data Insights

73% of AI group interacted with adaptive features, yet several said changes were "not obvious."

What Participants Said

"I did not see much change... If we are to adapt to each journey changing that main page is essential. If all of the changes happen in more hard-to-find areas it does not seem very adaptive." (GLG_26 – Survey Comment)

Recommendation

- Make adaptive changes **visually distinct** (color cues, dynamic layouts).
- Add **feedback prompts** when personalization occurs.

6

Recommendations and Next Steps

Strategic Priorities

Based on quantitative and qualitative findings across all objectives and comparative analysis, these recommendations focus on **feature refinement, UX improvements, and behavioral support** for the next development phase.

Double Down on AI Differentiators	• Enhance **Albert's conversational depth** (contextual guidance, proactive prompts). • Refine **photo-based meal logging** for accuracy and speed.
Personalization Engine	• Incorporate **dietary preferences, allergies, and behavioral patterns** into recommendations.
Adherence and Behavioral Support	• Expand **reminder system** for medication and logging tasks. • Integrate **CBT-informed coaching** in collaboration with behavioral specialists.
Mental Health Safeguards	• Shift from daily to **weekly weight tracking** to reduce stress and incorporate CBT into recommendations.

Footnote/Source

Q&A